

POWER PLANT ENGINEERING

A Power Plant is an industrial facility used to generate electricity (electric power) with the help of one or more generators, which converts different energy sources into electric power.

Types of power Plant

Types of power Plant for the generation of electric power include:

1. Thermal power Plant (Steam power Plant),
2. Hydro-electric power Plant,
3. Diesel power Plant,
4. Gas turbine power Plant,
5. Nuclear power Plant.

Steam power Plant

In steam power plants, the heat of combustion of fossil fuels (coal, oil or gas) is utilised by the boilers to raise steam at high pressure and temperature. The steam is then expanded in the prime mover (i.e., steam turbine) and is condensed in a condenser to be fed into the boiler again. Steam turbines used in steam power plants not only act as prime movers but also as drives for auxiliary equipment, such as pumps, — Stokes fans etc.

Steam power plant may be installed to either generate electrical energy only or to generate electrical energy along with the generation of steam for industrial purposes such as in refineries, sugar mills, textile mills, chemical works, food manufacturing etc.

Steam power plants may either be the condensing or the non-condensing type. In the condensing type power plants, the exhaust steam is discharged into a condenser, while in the non-condensing type power plants, the steam exhausted from the turbine is discharged at atmospheric pressure or at a pressure greater than atmospheric.

According to use, such plants can be classified into

- (i) Industrial power plants or Captive power plants and
- (ii) Central power plants.

Industrial power plants :- The industries requiring steam for process purposes may use steam turbines for generation of electrical energy also for its own use. The steam for process purposes may be tapped from extraction bleeding of the turbine at a pressure of $1.4 - 7 \text{ kg/cm}^2$. They are usually of the non-condensing type. Usually of small capacity (say up to 10 MW).

Central power plants :- Steam power plants are mainly used as central plants to generate electrical energy for supply to various consumers (industrial, agricultural, commercial, domestic etc.). They are usually of the condensing type. The size of the generating sets may be from 10-1,000 MW or even higher.

Steam power plant for the generation of electrical power is preferred where large amount of power is required to be generated and financial, climatic, and geographical conditions do not permit the installation of hydroelectric power plants. Also suitable where coal and water are in abundance.

Efficiency of Steam Power Plants

1. The thermal efficiency of Steam power plants, is defined as the ratio of heat equivalent of mechanical energy transmitted to the turbine shaft to the heat of combustion of coal. This is about 30%.

2. The overall efficiency of a Steam power plant is defined as the ratio of heat equivalent of electrical output to the heat of combustion of coal (about 29%).

Overall efficiency is less than the thermal efficiency because of losses (1%) that occur in the alternator.

Losses in Steam power plants

Losses occurring in Steam power plants may be summarized as follows:

1. Boiler house losses

(i) To dry flue gases	_____	5%
(ii) To moisture in gases	_____	5%
(iii) To ash and unburnt carbon	_____	1.0%
(iv) To radiation and leakage	_____	2.5%
(v) Unburnt losses	_____	2.5%
Total	_____	16%

2. Turbine losses

Heat rejected to condenser	_____	54%
Alternator losses	_____	1%
Total Output is about	_____	29%

Selection of Site for Steam/Thermal Power Plants

In order to achieve overall economy and efficient generation of electric power in the Steam power plant, the following factors need to be considered in the selection of a site:

- (i) Availability of water supply: Since large quantity of water is needed to raise the steam in boilers, for cooling purposes in Condensers, for the conveyance of ash disposal and for drinking purposes, the plant should be sited (located) near a river, sea, lake or near a canal to ensure continuous supply of water.
- (ii) Supply of fuel: The plant ~~should~~ be located near coal mines to minimise or avoid the transport of coal (fuel).

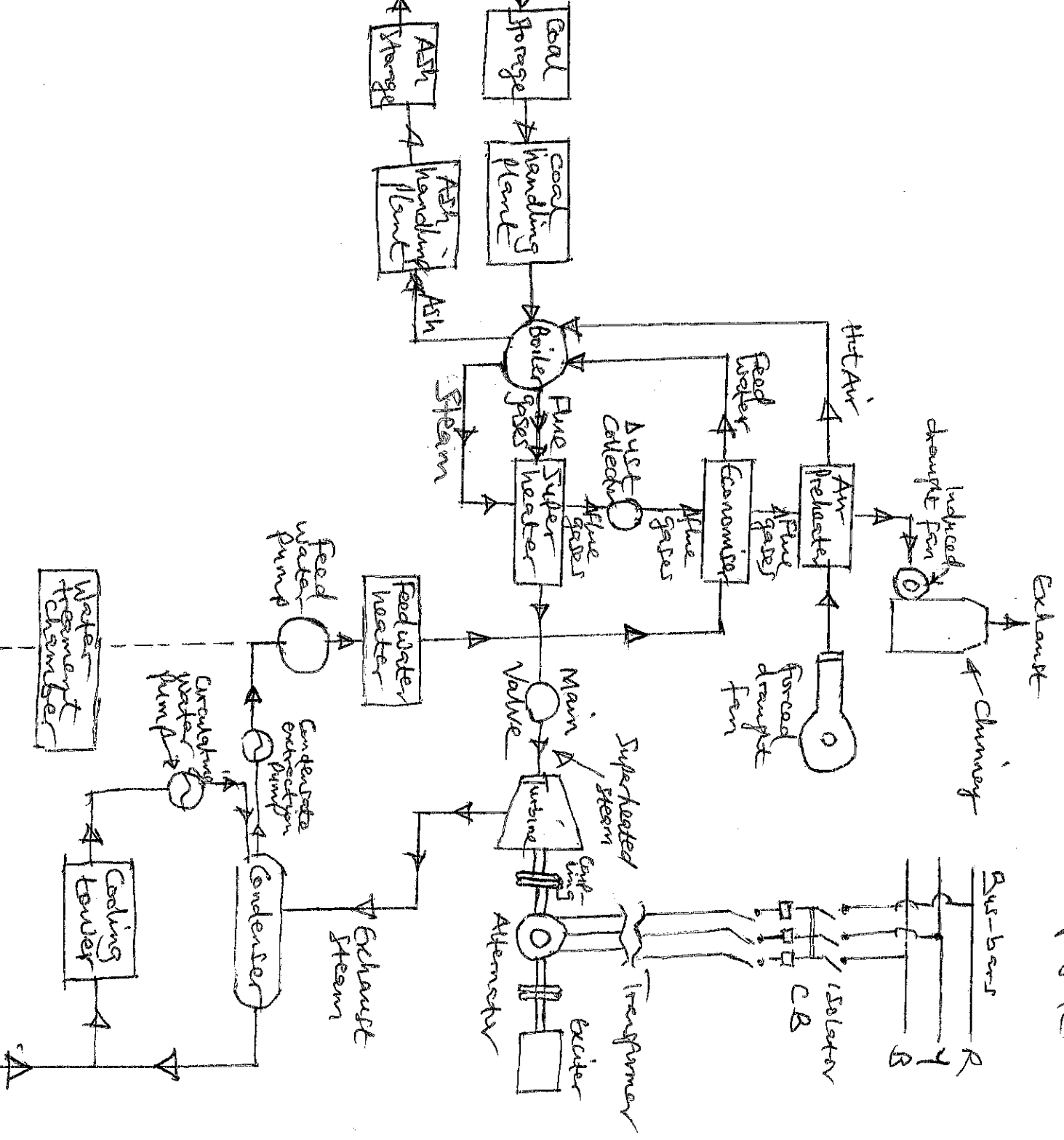
- (iii) Proximity to Load Centre: The power plant should be as near as possible to the centre of the load so that the transmission cost and losses are reduced. This factor is important when d.c. supply system is adopted. In the case of a.c. supply system when transmission of energy from lower voltage to higher voltage and vice versa is possible, power plants can be erected away from load centre provided other conditions are favourable.

- (iv) Transportation facilities: Transport facilities have to be readily available for the transportation of equipment, fuel and staff e.g., near railway station and/or connected to roads.

- (v) Supply of Labour: Skilled and unskilled labour should be available at reasonable rates near the location of the plant.

- (vi) Cost and type of land: The plant should be located at a place where land is cheap and further extension, if necessary, is possible. The bearing capacity of the ground should be adequate so that heavy equipment can be installed.

- (vii) Distance from populated area: As large amount of coal is burnt in a Steam power station, residues and fumes pollute the surrounding area. As such the plant should be located away from the populated areas.
- (viii) Ash disposal: Ash is the main waste product of the Steam power plant, hence some suitable means for the disposal of ash should be thought of in the selection of site.



Schematic arrangement of Steam power plant.

River

The Schematic arrangement of a modern Steam power plant is shown overleaf. The arrangement is divided into the following stages:

1. Coal and ash handling arrangement
2. Steam generating plant
3. Steam turbine
4. Alternator
5. Feed water
6. Cooling arrangement.

1. Coal and Ash handling Unit: The coal is transported to the power plant by road, rail or water and is stored in the coal storage plant. Coal storage is mainly a matter of protection against coal strikes, failure of transportation system and general coal shortages. From the coal storage plant, coal is delivered to the coal handling plant where it is pulverised (i.e., crushed into small pieces) in order to increase its surface exposure, thus promoting rapid combustion without using large quantity of excess air. The pulverised coal is fed to the boiler by belt conveyors. The coal is burnt in the boiler and the ash produced after the complete combustion of coal is removed to the ash handling plant and the ash is delivered to the ash storage plant for disposal. The removal of the ash from the boiler-furnace is necessary for proper burning of coal.

2. Steam generating plant: The steam generating plant consists of a boiler for the production of steam and other auxiliary equipment for the utilisation of flue gases.

(a) Boiler: The heat of combustion of coal in the boiler is utilized to convert water into steam at high temperature and pressure. The flue gases from the boiler make their journey through Super heater, economiser, air pre-heater and are finally exhausted to the atmosphere through the chimney.

(b) Superheater: The steam produced in the boiler is wet and is passed through a Superheater where it is dried and superheated (i.e., steam temperature increased above that of boiling point).

(4)

the flue gases on their way to chimney. Superheating provides two principal benefits, firstly, the overall efficiency is increased. Secondly, too much condensation in the last stages of turbine (which would cause blade corrosion) is avoided. The superheated steam from the superheater is fed to steam turbine through the main valve.

(c) Economiser. An economiser is essentially a feed water heater and derives heat from the flue gases for this purpose. The feed water is fed to the economiser before supplying to the boiler. The feed water extracts a part of heat of flue gases to increase the feed water temperature.

(d) Air preheater. An air preheater increases the temperature of the air supplied for coal burning by deriving heat from flue gases. Air is drawn from the atmosphere by a forced draught fan and is passed through air preheater before supplying to the boiler furnace. The air preheater extracts heat from flue gases and increases the temperature of air used for coal combustion. The principal benefits of preheating the air are: increased thermal efficiency and increased steam capacity per square metre of boiler surface.

3. Steam turbine: The dry and superheated steam from the superheater is fed to the steam turbine through main valve. The converted into mechanical energy. After giving heat energy to the turbine, the steam is exhausted to the condenser which condenses the exhaust steam by means of cold water circulation.

4. Alternator: The steam turbine is coupled to an alternator. The alternator converts the mechanical energy of turbine into electrical energy. The electrical output from the alternator is delivered to the bus bars through transformer, circuit breakers and isolators.

5. Feed Water: The condensate from the condenser is used as feed water to the boiler. Some water may be lost in the cycle which is suitably made up from external source. The feed water on its P.T.O

way to the boiler is heated by water headers and economiser. This helps in raising the overall efficiency of the plant.

6. Cooling arrangement: In order to improve the efficiency of the plant, the steam exhausted from the turbine is condensed by means of a condenser. Water is drawn from a natural source of supply such as a river, canal or lake and is circulated through the condenser. The circulating water takes up the heat of the exhausted steam and itself becomes hot. This hot water coming out of the condenser is discharged from the source of supply into the river. In case the availability of water towers are used. During the scarcity of water in the river, hot water from the condenser is passed on to the cooling towers where it is cooled. The cold water from the cooling tower is reused in the condenser.

Advantages of Steam power plants

The steam power plants have the following merits over other plants:

- (i) The fuel (i.e., coal) used is cheaper.
- (ii) Less initial cost as compared to other generating plants of the same capacity.
- (iii) It can be installed at any place irrespective of existence of coal. The coal can be transported to the site of the plant by rail or road.
- (iv) It requires less space as compared to the hydroelectric power plant.
- (v) Cheaper in production cost in comparison with that of diesel power plant.
- (vi) Able to respond to rapidly changing loads without difficulty.
- (vii) Steam engines and turbines can work under 25% of overload continuously.
- (viii) A portion of steam raised can be used as process steam in various industries (paper mills, textile mills, sugar mills, refineries, chemical works etc.).

Disadvantages of Steam Power Plants

The Steam Power Plants possesses the following disadvantages:

- (i) Costlier in operating cost in comparison with that of hydro- and nuclear power plants.
- (ii) Requires long time for erection and put into action.
- (iii) Efficiency falls rapidly below 75% of full load.
- (iv) High maintenance cost.
- (v) Requirement of water in huge quantity.
- (vi) Pollution of atmosphere due to fumes and residues from put-
terised fuels.
- (vii) Troubles from smoke and heat from the plant.
- (viii) Handling of coal and disposal of ash is quite difficult.

The Boiler

The boiler or steam generator, is a closed vessel made of steel. Its function is to transfer the heat produced by the combustion of fuel (solid, liquid or gaseous) to water and utilizing it to generate steam.

Some important terms used for steam boilers are:

1. Boiler Shell :- It is made up of steel plates bent into

cylindrical form and riveted or welded together. The ends of the shell are closed by means of end plates. It has sufficient capacity to contain water and steam.

2. Grate :- It is a platform in the combustion chamber —

where fuel (coal or wood) is burnt. It generally consists of cast iron bars which are spaced apart so that air (needed for combustion) can pass through them.

3. Furnace :- Is the space above the grate and below the boiler shell, in which the fuel is actually burnt. It is also called the fire box.

4. Boiler Mountings:- These are the fittings which are mounted on the boiler for its proper functioning. They include water level indicator, pressure gauge, Safety valve etc. Boilers can not function safely without the mountings.

5. Boiler Accessories:- These are devices which form an integral part of the boiler but are not mounted on it. They include the Super heater, economiser, feed pump, air pre heater etc. They help in running the boiler efficiently.

Essentials of a good Steam Boiler

A good Steam boiler should possess the following essentials:

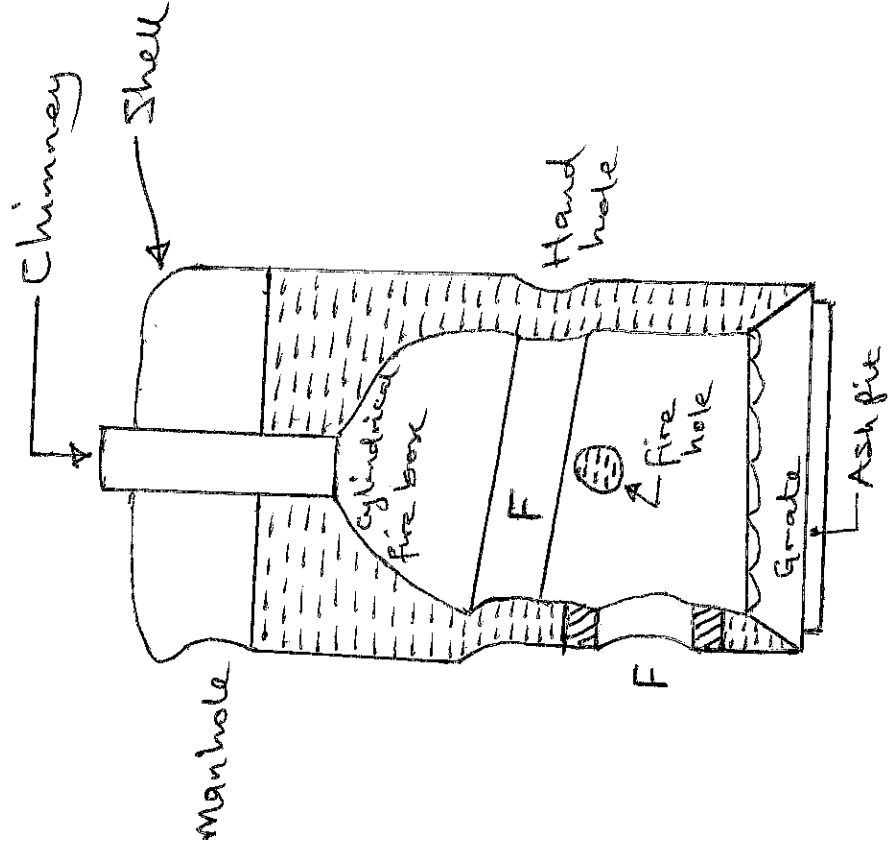
1. It should produce maximum quantity of steam with minimum fuel consumption.
2. It should be economical to install, and should require little attention during operation.
3. It should be able to rapidly meet load fluctuation(s).
4. It should be capable of quick starting.
5. It should be light in weight.
6. It should occupy a small space.
7. It should have few joints and accessible for inspection.
8. The mud and other deposits should not collect on the heating plates.
9. The tubes should not accumulate soot or water deposits and should have a reasonable margin of strength to allow for wear and corrosion.
10. The water and flue gas circuits should be designed to allow a maximum fluid velocity without incurring heavy frictional loss.

Classification of Steam Boilers

Boilers are generally classified in the following ways:

1. According to the contents in the tube;
 - (a) Fire tube or smoke tube boiler, and
 - (b) Water tube boiler.
2. According to the position of the furnace;
 - (a) Internally-fired boilers, and
 - (b) Externally-fired boilers.
3. According to the axis of the shell;
 - (a) Vertical boilers, and
 - (b) Horizontal boilers.
4. According to the number of tubes;
 - (a) Single tube boilers, and
 - (b) Multitubular boilers.
5. According to the source of heat, these sources are;
 - (a) From combustion of solid, liquid or gaseous fuels.
 - (b) From hot waste gases from chemical processes.
 - (c) From electrical energy or
 - (d) From Nuclear energy.
6. According to the method of circulation of water and steam;
 - (a) Natural circulation boilers, and
 - (b) Forced circulation boilers.
7. According to the use;
 - (a) Stationary boilers, and
 - (b) Mobile boilers.

The Simple Vertical Boiler



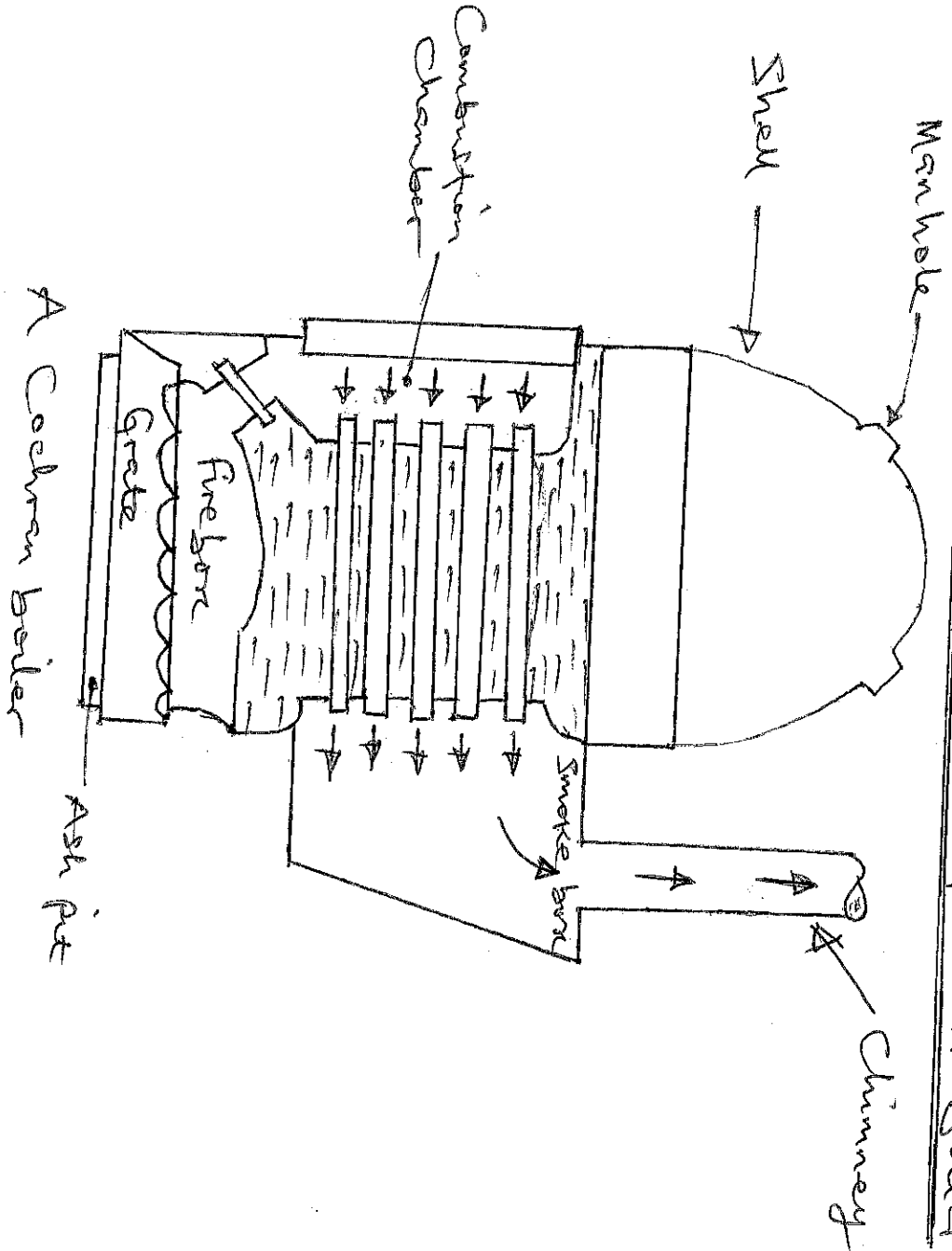
A Simple Vertical Boiler

A Simple Vertical boiler produces steam at a low pressure and in small quantities. It is used for low power generation or at places where the space is limited. The boiler consists of a cylindrical shell surrounding a nearly cylindrical firebox. The firebox is slightly tapered towards the top to allow the ready passage of the steam to the surface.

At the bottom of the firebox is the grate. The firebox is fitted with two or more inclined cross tubes F, F. The inclination is provided to increase the heating surface as well as to improve the circulation of water. An uptake tube passes from the top of the fire tube to the chimney. The hand holes are provided opposite to the end of each water tube for cleaning deposits. A manhole is provided at the top for a man to enter and clean the boiler.

A mudhole is provided at the bottom of the shell to remove the mud that settles down. The space between the boiler shell and fire box is filled with water to be heated.

Cochran Boiler or Vertical Multitubular Boiler



A Cochran boiler is considered to be one of the vertical multitubular boilers. It is an improved type of a single vertical boiler.

The boiler consists of an external cylindrical shell and a fire box as shown in the figure above. The shell and fire box are both hemispherical. The hemispherical crown of the boiler shell gives maximum space and strength to withstand the pressure of steam inside the boiler. The hemispherical crown of the fire tube is an advantage for resisting intense heat. The fire box and combustion chamber are connected through a short pipe. The gases from the combustion chamber flow to the smoke box through a number of smoke tubes. The gases from the smoke box pass to the atmosphere through a chimney.

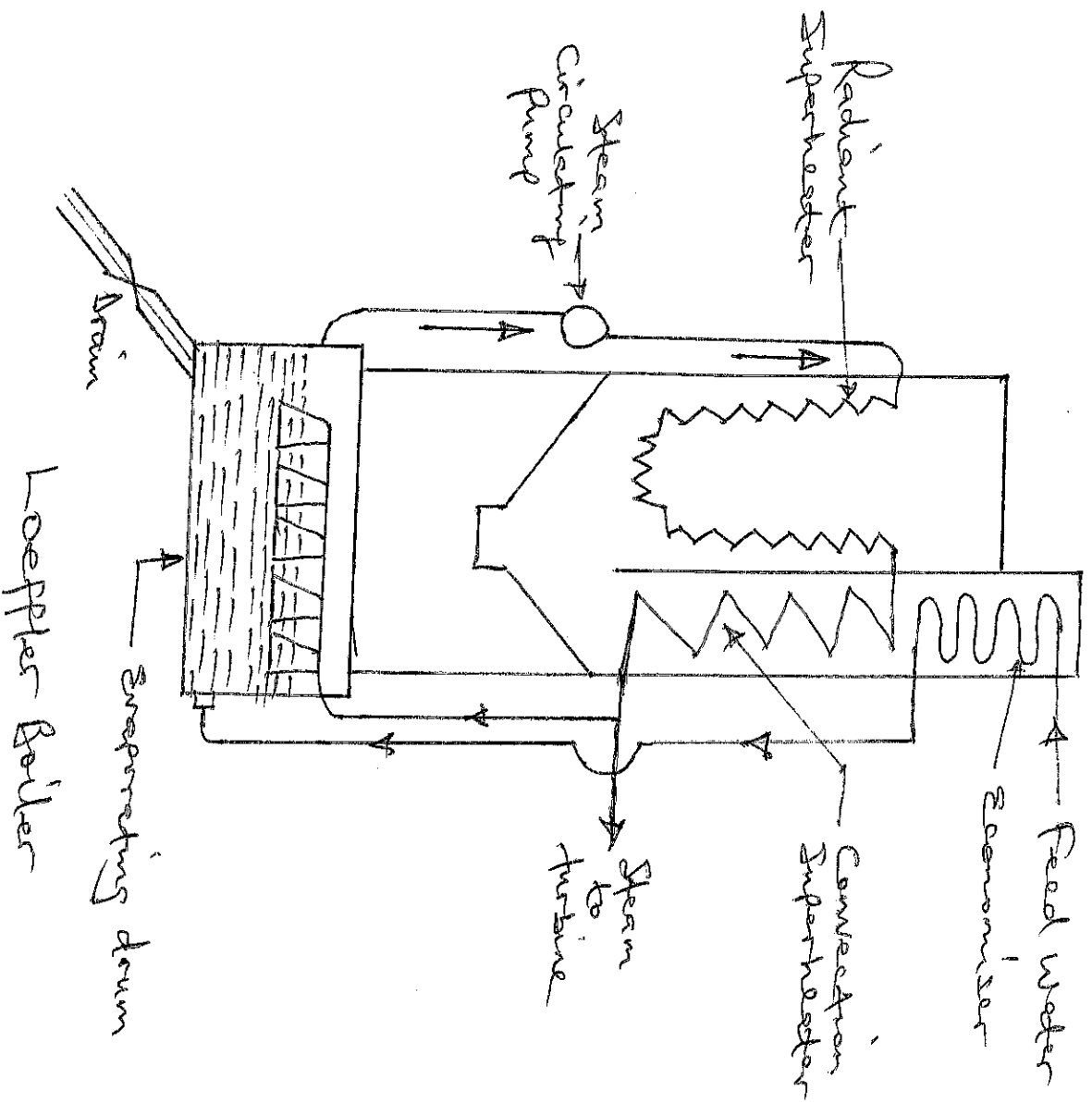
The combustion chamber is lined with fire bricks on the shell side. A man hole near the top of the crown on the shell is provided for cleaning.

At the bottom of the firebox, there is a grate (in case of coal firing) and the coal is fed through the fire hole. If the boiler is used for oil firing, no grate is provided, but the bottom of the firebox is lined with fire bricks. The oil burner is fitted at the fire hole.

Loeffler Boiler

This is a water tube boiler using a forced circulation. Its main principle of working is to evaporate the feed water by means of superheated steam from the superheater. The hot gases from the furnace are used for superheating. The feed water from the economiser tubes is forced to mix with the superheated steam in the evaporating drum. The saturated steam, thus formed, is drawn from the evaporating drum by a steam circulating pump. This steam passes through the tubes of the combustion chamber walls and then enters the superheater. From the superheater, about one-third of the superheated steam passes to the turbine and the remaining two-thirds is used to evaporate the feed water in the evaporating drum.

(8)



Evaporative Capacity :- The evaporative capacity or power of a boiler is the amount of water evaporated or steam produced in kg/h. It may also be expressed in kg/kg of fuel burnt or kg/h/m² of heating surface.

Equivalent Evaporation :- It is the amount of water evaporated from feed water at 100°C and formed into dry and saturated steam at 100°C at normal atmospheric pressure. It is usually written as "from and at 100°C".

Since the water is already at the boiling temperature, it requires only latent heat at 1.013 bar to convert it into steam at the temperature (100°C). The value of this latent heat is taken as 2257 kJ/kg. P.T.O

Mathematically,

Equivalent evaporation "from and at 100°C ".

$$E = \frac{\text{Total heat required to evaporate feed water}}{2257}$$

$$= \frac{m_e (h - h_{f1})}{2257}$$

or $\frac{h - h_{f1}}{2257}$ is known as factor of evaporation (F_e).

m_e = mass of water actually evaporated or steam produced in kg/h or kg/kg of fuel burnt.

$h - h_{f1}$ = heat required to evaporate 1 kg of water
 t_1 = temp. of feed water in $^{\circ}\text{C}$.

h = enthalpy or total heat of steam in kg/kg of steam corresponding to a given working pressure (from steam tables)
 $h = h_f + x h_{fg}$ --- (for wet steam)

$$= h_f + h_{fg} = h_g \quad (\text{for dry sat. steam})$$

$$= h_g + c_p (t_{\text{sup}} - t) \quad \text{--- (for superheated steam)}$$

Boiler Efficiency :- Is defined as the ratio of heat actually used in producing the steam to the heat liberated in the furnace. It is known as thermal efficiency of the boiler.

$$\text{Boiler efficiency or thermal efficiency } (\eta) = \frac{\text{Heat actually used in prod. using steam}}{\text{Heat liberated in the furnace}} \\ = \frac{m_e (h - h_{f1})}{C}$$

where C = calorific value in kg/kg of fuel

If m_s = total mass of water evaporated into steam in kg and m_f = mass of fuel used in kg

$$\text{then } m_e = \frac{m_s}{m_f} \text{ kg/kg of fuel}$$

$$\text{and } \eta = \frac{m_s (h - h_{f1})}{m_f \times C}$$

Fuel

A fuel may be defined as a substance (containing mostly Carbon and Hydrogen) which on burning with oxygen in the atmosphere, produces a large amount of heat.

Classification of Fuels

Fuel may be classified as:

1. Solid fuels,
2. Liquid fuels, and
3. Gaseous fuels.

Each of these fuels may be further subdivided into:

- (a) Natural fuels, and
- (b) Prepared fuels.

The various fuels commonly used for combustion in the thermal power plants are coal, oil and gas. Gaseous fuel is rarely economical except when the power plant is located near natural gas field or gas manufacturing industries. Oil is used only where it is plentiful and cheap. Coal is the most commonly used fuel in thermal power plants.

Solid Fuels

The natural solid fuels are wood, peat, lignite or brown coal, bituminous coal and anthracite coal. The prepared solid fuels are wood charcoal, coke, briquetted coal and pulverised coal.

1. Wood:- At one time extensively used as fuel. It consists mainly of carbon and hydrogen. It is converted into coal when burnt in the absence of air.

2. Peat:- Is a spongy humid substance found in boggy land (wet and muddy land). It is regarded as the first stage in the process of transformation of buried vegetation into coal.

P.T.O

It has large amount of water contents (moisture contents) and low Carbon content. Not suitable for use in power plant.

3. Lignite:- It is the next stage in the development of coal after peat. It has nearly 40% moisture and 60% of Carbon. It can be used for proper generation.

4. Bituminous Coal:- Is the next stage after lignite in coal formation and have low moisture content (4-6%) and 75-90% of Carbon. It is weather resistant and burns with a yellow flame. It is the most commonly used coal and widely utilized for almost all purposes including power generation.

5. Anthracite Coal:- This is the final stage in the coal formation, and contains 90% or more Carbon with a very little volatile matter. It is comparatively smokeless and has little flame. Is valuable for steam raising and general power purposes.

6. Wood Charcoal:- It is made by heating wood with a limited supply of air to a temperature not less than 280°C . It is a good prepared fuel, and is used for various metallurgical processes.

7. Coke:- Is produced when coal is strongly heated continuously for 42 to 48 hours in the absence of air in a closed vessel. This process is known as carbonisation of coal. It is dull black in colour, porous and smokeless. It has about 80-90% Carbon content.

8. Briquetted Coal:- Is produced from finely ground coal by moulding under pressure with or without a binding material (Coal tar, clay etc.)

9. Pulverised Coal:- This is the low grade coal with a high ash content, is powdered to produce pulverised coal. It is first dried, then crushed into a fine powder by a pulverising machine. Widely used in cement industries and in metallurgical processes.

NB Among the solid fuels, anthracite coal is commonly used in all types of heat engines.

Liquid Fuels

Oil can be used in a boiler furnace for generating steam. The natural petroleum may be separated into petrol or gasoline, paraffin oil or kerosene, fuel oils and lubricating oils.

The liquid fuels (heavy fuel oils) distilled after petrol and kerosene, are used in diesel engines and in oil fired boilers.

Advantages of Liquid fuels over Solid fuels

The following are some of the merits of liquid fuels

Over Solid fuels:

1. Higher Calorific Value.
2. Lower Storage Capacity required.
3. Better economy in handling.
4. Better control of consumption by using valves.
5. Better cleanliness and freedom from dust.
6. Practically no stress.
7. Non-deterioration in storage.
8. Non-corrosion of boiler parts.
9. Higher efficiency.

Disadvantages of Liquid fuels Over Solid fuels

The following are some of the disadvantages of liquid fuels over solid fuels:

1. Higher cost.
2. Greater risk of fire.
3. Costly containers are required for storage and transport.

P.T.O

Gaseous Fuels

The gaseous fuel may either be natural or prepared/manufactured. The latter being Cool gas. Usually, the natural gas is found in or near the petroleum fields under the earth's surface and usually used for power generation. The natural gas essentially consists of Marsh gas or methane (CH_4) together with small amount of other gases such as C_2H_6 , CO_2 and CO .

Examples of prepared gas used as fuel are:

- (a) Coal gas, (b) producer gas, (c) Water gas, (d) Mond gas, (e) Blast furnace gas, (f) Coke oven gas.

Gaseous fuels have all the advantages of oil/liquid fuels except for ease of storage. The major limitation for using natural gas as fuel for power generation is location of power plant near the natural gas fields otherwise the transportation cost will be high.

Assignment

Write Short notes on:

- (1) Coal gas,
- (2) producer gas,
- (3) Water gas,
- (4) Mond gas,
- (5) Blast furnace gas, and
- (6) Coke oven gas.

Firing Steam Boiler With Coal

There are mainly two methods of firing steam boilers with coal, namely:

1. Solid fuel firing, and
2. Pulverized fuel firing.

In solid fuel firing, we have

- (1) Hand firing, and
- (2) Stoker or mechanical firing.

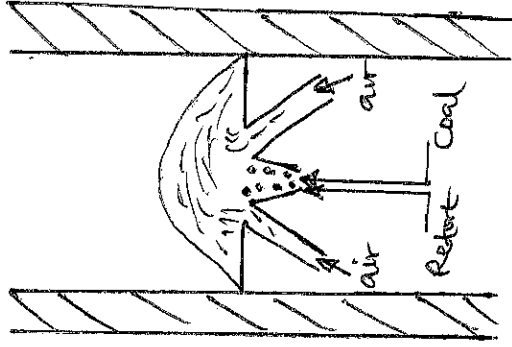
Hand firing:- Here, the coal is put into the furnace frequently by shovels. The air needed for combustion is regulated by dampers. The grate of such boilers consists of bars over which the coal is put. This method can be used only in small installations as uniformity of combustion is difficult to control. It was used in the past to drive coal engine locomotives. Adjustments for the air supply need to be made everytime coal is fed to the furnace.

Stoker firing:- This is obtained when coal is put into the steam boiler furnace by means of a mechanical stoker. Mechanical stokers receive fuel and then carry it to the furnace for combustion and after combustion discharge the ash at the appropriate point. In this method, there is uniform feed of fuel into the furnace, easy control of firing, saving in labour cost, fluctuation in load demands can be met because of control of combustion etc.

There are mainly two types of mechanical stokers:

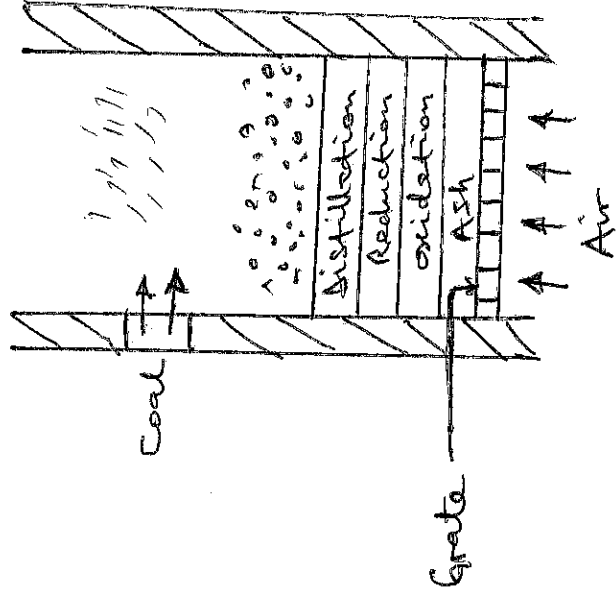
- (i) Under-feed stoker firing, and
- (ii) Over-feed stoker (chain grate stokers, travelling grate stokers and spreader stokers).

Under-feed Stokers :- In under-feed Stokers, fuel (Coal) is admitted into the furnace below the point of air admission. Coal in the hopper is pushed into the retort by means of a reciprocating plunger. They are suitable for burning Bituminous Coal. Air is admitted into the furnace through holes in the retort side.



Under-feed Stoker.

Over-feed Stokers :- In Overfeed Stokers, the coal is supplied into the grate above the point of admission of air.



Overfeed Stoker.

Pulverised fuel firing :- This is most modern and efficient method of boiler firing. Solid fuels can be used in a powdered form and burn like oil and gases. The coal is first dried usually by the gases, ground to a fine powder in a pulverised mill and then projected into the combustion chamber by means of a current of hot air. A volume of preheated air to make up the necessary amount for combustion is blown in separately, and the resulting turbulence in the high pressure combustion chamber helps in thorough combustion of the fuel. The air used to dry the coal and convey the powdered fuel to the furnace is called the primary air and the air blown separately to complete the combustion is called the secondary air.

Assignment

Describe with sketches the firing systems for —
Liquid and gaseous fuels.

